

DOCKER

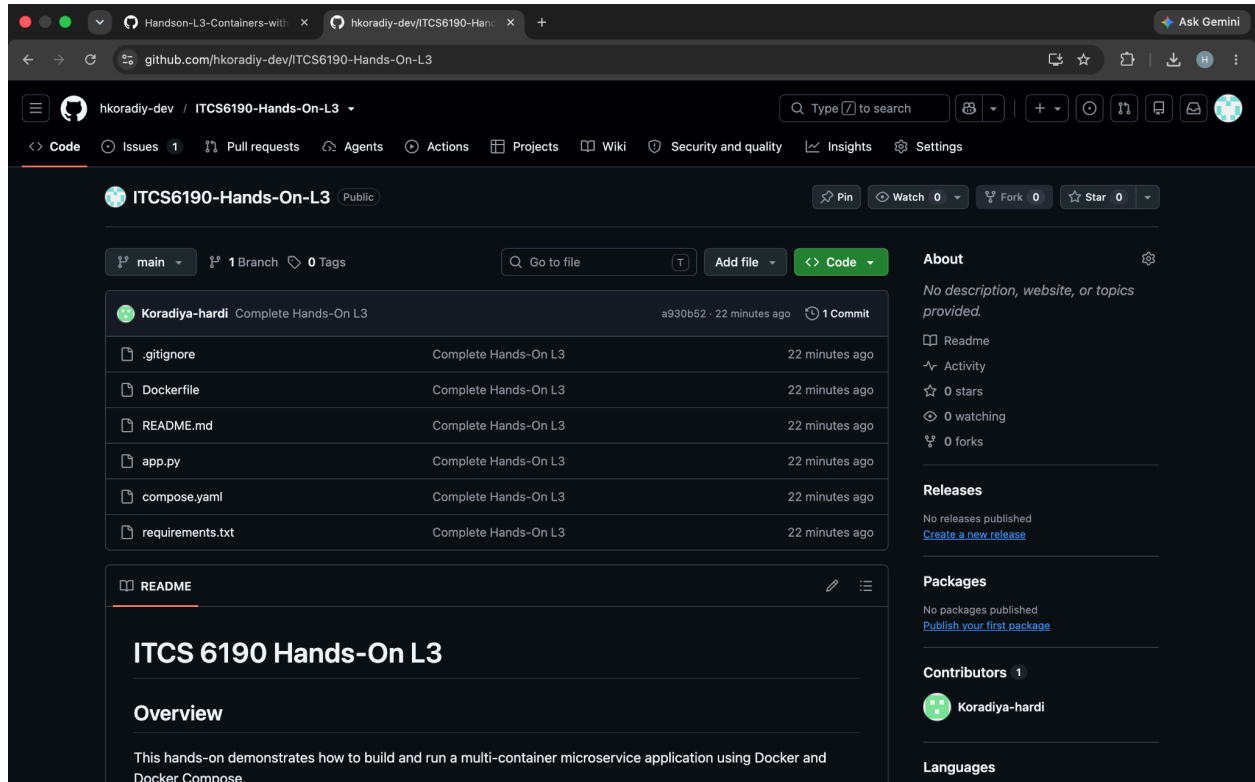
1. Docker and Application Setup

- Docker was verified successfully using `docker --version`. PostgreSQL 16 was pulled and run in a Docker container named `postgres1`.
- The hands-on application was then configured with Python Flask and Redis and orchestrated using Docker Compose. The application was
- tested at `http://localhost:8000`, where the Redis-backed hit counter increased after refreshing the page.

```
1ed0bcd2bfcd1adcf94d0f1beb4dc7d3fa1c90bbebdadce4068393344c0b3747
ERR[0000] error waiting for container:
(base) hardikoradiya@hardis-MacBook-Air ~ % docker ps -a
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS                                NAMES
1ed0bcd2bfcd   postgres:16                         "docker-entrypoint.s..." 25 seconds ago Up 24 seconds 0.0.0.0:5432->5432/tcp postgres1
b5102bcb6ba6   selenium/hub:latest                "docker-entrypoint.s..." 18 months ago Exited (143) 18 months ago selenium-hub
6ada9eeb6ba2   nodejs-app                          "docker-entrypoint.s..." 2 years ago   Exited (137) 2 years ago beautiful_nobel
a06228cf8447   nodejs-app                          "docker-entrypoint.s..." 2 years ago   Exited (137) 2 years ago bold_pare
4460e73a4d0e   28652978f269                       "docker-entrypoint.s..." 2 years ago   Exited (137) 2 years ago zealous_tesla
b0cae58ec3dd   2848-game-2                         "docker-entrypoint.s..." 2 years ago   Exited (0) 2 years ago gallant_chaum
5a73bc9338a1   285512a7f31d                       "docker-entrypoint.s..." 2 years ago   Exited (0) 2 years ago ecstatic_lovelace
7a31210da86a   5cfa723c6206                       "docker-entrypoint.s..." 2 years ago   Exited (127) 2 years ago sharp_lederberg
b65f5845103a   b84cf4bd4ec2                       "docker-entrypoint.s..." 2 years ago   Exited (1) 2 years ago kind_khayyam
53bf733444f4   5cfa723c6206                       "docker-entrypoint.s..." 2 years ago   Exited (127) 2 years ago interesting_khorana
34fec62d190e   prac8-web                           "docker-entrypoint.s..." 2 years ago   Created                                             web
3118ba41e123   redislabs/redismod                 "docker-entrypoint.s..." 2 years ago   Exited (137) 2 years ago redis
02a1800aa504   hardipatel15/mern-app              "docker-entrypoint.s..." 2 years ago   Created                                             eloquent_germain
8a73cf604c33   hardipatel15/mern-app              "docker-entrypoint.s..." 2 years ago   Exited (1) 2 years ago loving_hypatia
4793a826ac19   hardipatel15/react-app              "docker-entrypoint.s..." 2 years ago   Exited (137) 2 years ago affectionate_wilbur
797cb8e65a4a   gcr.io/k8s-minikube/kicbase:v0.0.42 "docker-entrypoint.s..." 2 years ago   Exited (138) 2 years ago minikube
7179e602eebf   my-img                             "docker-entrypoint.s..." 2 years ago   Exited (137) 2 years ago distracted_cartwright
f70e9d19d2cd   my-img                             "docker-entrypoint.s..." 2 years ago   Exited (137) 2 years ago youthful_gagarin
81712ed1dced   my-img                             "docker-entrypoint.s..." 2 years ago   Exited (137) 2 years ago sweet_ellis
080cef280ad7   48efc6efe6ed                       "docker-entrypoint.s..." 2 years ago   Exited (137) 2 years ago infallible_keldysh
7977cb7d0471   48efc6efe6ed                       "docker-entrypoint.s..." 2 years ago   Exited (137) 2 years ago amazing_heyrovsky
89eb477664c5   my-first-img                       "docker-entrypoint.s..." 2 years ago   Exited (137) 2 years ago eloquent_allen
f1b8aa5e5bf1   my-first-img                       "docker-entrypoint.s..." 2 years ago   Exited (137) 2 years ago agitated_perlman
b538d2dd38fa   d87baa23eddb                       "docker-entrypoint.s..." 2 years ago   Exited (137) 2 years ago kind_clarke
7921b9c30d23   d87baa23eddb                       "docker-entrypoint.s..." 2 years ago   Exited (137) 2 years ago strange_tu
ad91f838699c   hello-world                         "docker-entrypoint.s..." 2 years ago   Exited (0) 2 years ago stupefied_haslett
b593b0130672   test_app                           "docker-entrypoint.s..." 2 years ago   Exited (137) 2 years ago backend2
7a80d21f4faf   mongo                              "docker-entrypoint.s..." 2 years ago   Exited (0) 2 years ago unruffled_einstein
579c87436b24   test_app                           "docker-entrypoint.s..." 2 years ago   Exited (137) 2 years ago condescending_dubinsky
91b3fc978f68   test_app                           "docker-entrypoint.s..." 2 years ago   Exited (255) 2 years ago keen_einstein
5220bb8bf5c3   hello-world                         "docker-entrypoint.s..." 2 years ago   Exited (0) 2 years ago brave_leakey
(base) hardikoradiya@hardis-MacBook-Air ~ % docker exec -it postgres1 bash
[root@1ed0bcd2bfcd:/# psql -d postgres -U postgres
psql (16.15 (Debian 16.15-1.pgdg13+2))
Type "help" for help.

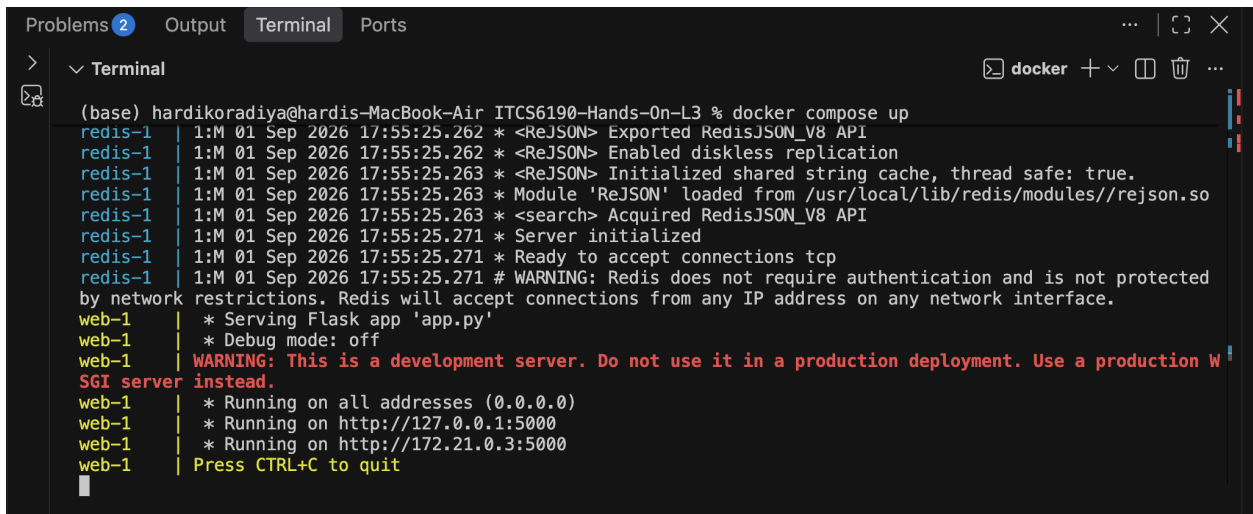
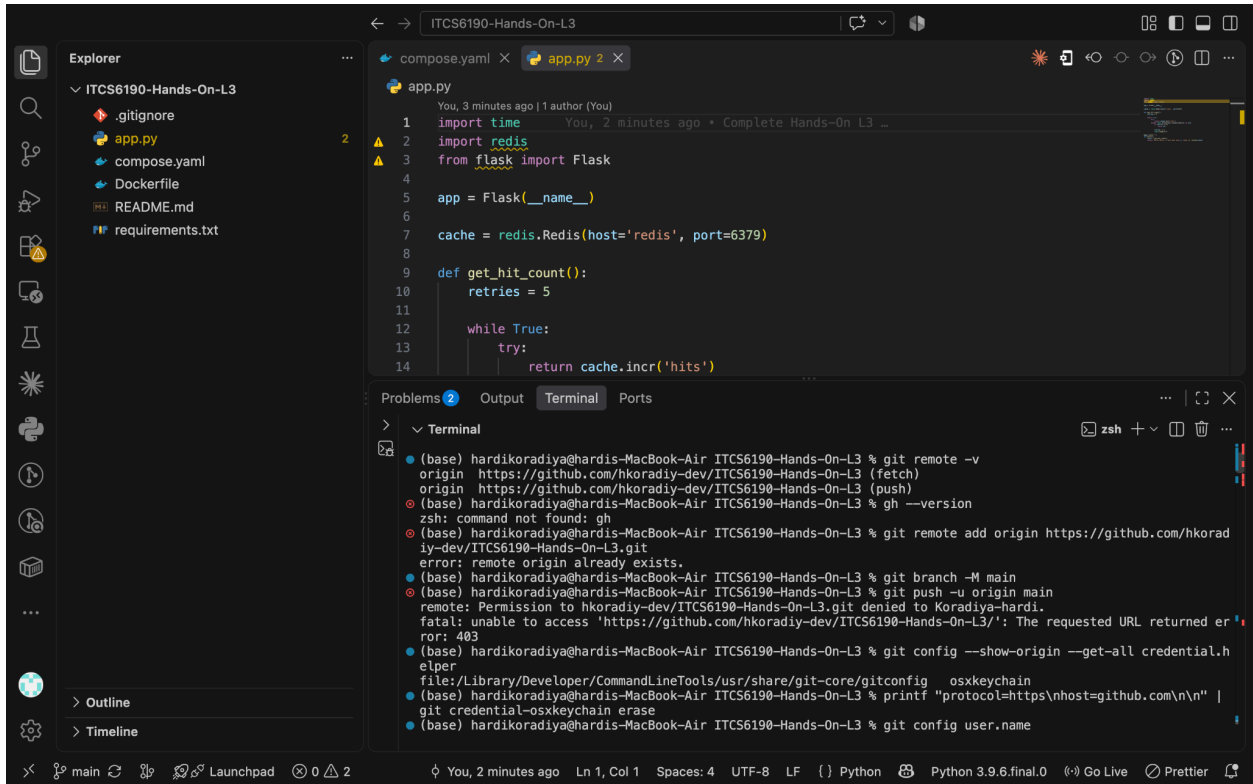
postgres=# \l
               List of databases
  Name | Owner  | Encoding | Locale Provider | Collate | Ctype  | ICU Locale | ICU Rules | Access privileges
-----+-----+-----+-----+-----+-----+-----+-----+-----
 postgres | postgres | UTF8     | libc             | en_US.utf8 | en_US.utf8 |             |           | =c/postgres
 template0 | postgres | UTF8     | libc             | en_US.utf8 | en_US.utf8 |             |           | =c/postgres
 template1 | postgres | UTF8     | libc             | en_US.utf8 | en_US.utf8 |             |           | =c/postgres
(3 rows)

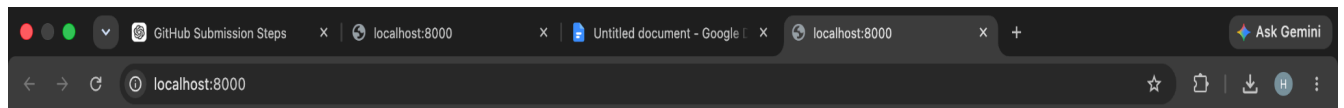
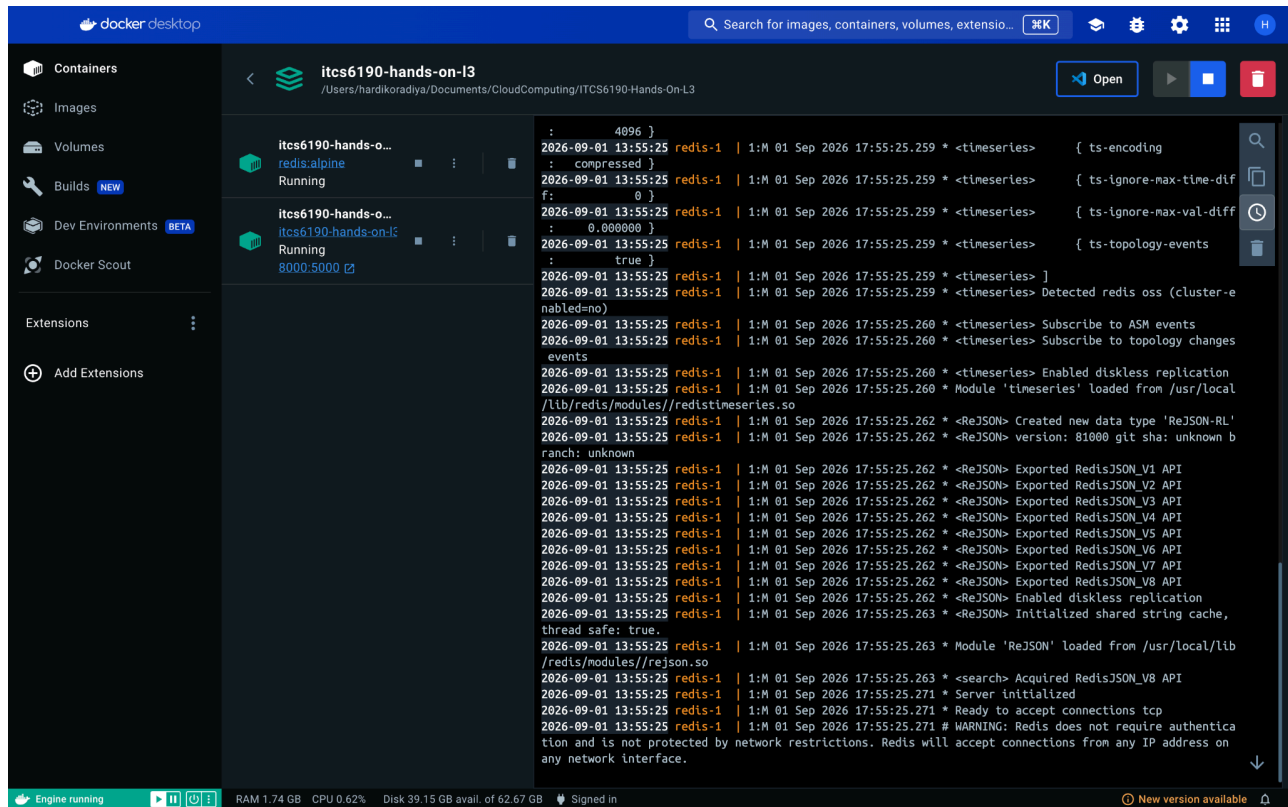
postgres=#
```



Errors Encountered:-

- **Docker wait message:** Docker displayed ERRO[0000] error waiting for container: when starting PostgreSQL. Checking docker ps -a showed that postgres1 was actually Up and mapped to port 5432, so no additional fix was required.
- **GitHub Permission Denied (403):** A push was rejected because GitHub authenticated the request as Koradiya-hardi while the repository belonged to hkoradiy-dev. The repository remote was verified and the Git credential configuration was investigated. The issue was resolved by using the correct GitHub account authentication.
- **Remote origin already exists:** Git reported that the origin remote already existed when it was added again. The existing remote was verified with git remote-v; no new remote was necessary.





What I Learned

This hands-on helped me understand how Docker containers can be used to package and run applications consistently. I learned how to create a Dockerfile, install application dependencies, create multiple services using Docker Compose, and connect a Flask application with Redis. I also learned how Docker Compose simplifies the management of multiple containers and how services can communicate using service names.

Submission LINK :-

<https://github.com/hkoradiy-dev/ITCS6190-Hands-On-L3>

